

Vishal Kumar Allam

Chicago, IL | 816-666-2813 | itsmevishal36@gmail.com | **H1B** | [Linkedin](#)

PROFESSIONAL SUMMARY

- IT Professional with over **5+ years** of experience in **quality engineering** and **technical integration testing** across Agile teams, validating backend systems, **RESTful APIs**, and cloud-native platforms for enterprise and **healthcare** customers.
- Hands-on experience validating **AWS** cloud services including **AWS Lambda**, **API Gateway**, **S3**, **ECS**, and **EventBridge**-driven **event-driven architecture** workflows as part of cloud-native application testing for healthcare platforms.
- Skilled in testing **API integration and authentication** flows including **RESTful APIs**, **OAuth2/OIDC**, **JWT**, **SAML 2.0**, **webhooks**, and **mTLS**-secured service-to-service communication using **RESTAssured** and Postman.
- Experience validating **healthcare data** exchange and **payer side** workflows at CVS Health Care Systems, including **prior authorization**, **utilization management**, claims and eligibility data governed by **HIPAA** and healthcare regulatory requirements.
- Working knowledge of **data architecture** and **integration patterns**, including **data modeling**, **schema design**, **transformation logic**, and **message queue/middleware**-based system validation.
- Experienced Quality Engineer with a strong background ensuring quality and reliability of software products, utilizing a diverse tech stack including **Java**, **Python**, **C#**, **TypeScript**, **.NET**, **Node.js**, **Playwright**, and **AWS Lambda**.
- Automation and Manual Testing: Extensive hands-on experience with **Selenium WebDriver**, **Cucumber**, and **Appium**, contributing significantly to product health and **production deployment** readiness.
- Database expertise across **SQL** and **NoSQL** systems including **PostgreSQL**, **MySQL**, **MongoDB**, **DynamoDB**, **Oracle 11g**, and **Vertica**, with focus on **query optimization** and data validation.
- Familiarity with **Identity and Access Management (IAM)** concepts including **SSO**, **RBAC**, **federated identity**, and **multi-tenancy** security validation for enterprise platforms.
- Frameworks and Methodologies: Proficient in using **Page Object Model**, **BDD Framework**, **Agile Methodology**, and **Scrum** teams to deliver on ambiguous, fast-moving requirements with high agency.
- Implementation of **Playwright**-based test framework with **TypeScript** for web application and **API** testing across production-grade services.
- Exposure to **Infrastructure-as-Code** concepts including **Terraform** and **CloudFormation** while validating environment provisioning for **cloud-native** test environments.
- Mobile Testing: Well-versed with Mobile Native Application and Mobile Web Testing using MAC Computers, iOS and Android Mobile Native Application testing using **Appium**.
- Tools and Platforms: Experienced with **WebStorm**, **Eclipse**, **GitHub**, **Jenkins**, **Confluence**, **QTP**, **Slack**, **Maven**, **TeamCity**, **JIRA**, and **Postman**.
- Test Case Development: Achieved **90% test case development efficiency** with a modern Test Case architecture covering **regulated environments** such as **HIPAA** and **SOC2**.
- Agile and CI/CD: Active participation in **Agile projects**, **Scrum** teams, and **CI/CD pipeline** tasks, enhancing overall **production deployment** efficiency and cross-functional collaboration with Engineering and Product teams.
- Certifications: **Oracle Certified Associate**, **ISTQB Certified Professional**, **Microsoft Technology Associate** in Database Administration Fundamentals and HTML-5 Application Development Fundamentals.

EDUCATION

University of Missouri Kansas City

"Master of Science" in Computer Science

Kansas City, MO

Jan. 2023 – May 2024

Jawaharlal Nehru Technological University

"Bachelor of Technology" in Computer Science

Hyderabad, India

Aug. 2015 - May 2019

TECHNICAL SKILLS

- **Cloud & AWS:** AWS Lambda, API Gateway, Amazon S3, Amazon ECS, Amazon RDS, EventBridge, Cloud-Native Application Testing.
- **API Integration & Authentication:** RESTful APIs, REST Assured, OAuth2, OIDC, SAML 2.0, JWT, mTLS, Webhooks, API Key Management, Postman, SoapUI.
- **Healthcare Data & Compliance:** HL7v2, FHIR, X12 (837, 278, 275), HIPAA, SOC2, HITRUST, Payer Side Healthcare Workflows, Health Plans, Managed Care Organizations, Prior Authorization, Utilization Management, Claims Processing.
- **Programming Languages:** C#, Python, Java, J2EE, SQL, Node.js.
- **Testing Frameworks:** BDD, XUnit, POM, Data Driven, Rest Assured, Keyword Driven, Playwright, Cucumber, JUnit, TestNG.
- **Databases & Data Modeling:** PostgreSQL, MySQL, Oracle, MongoDB, DynamoDB, Data Modeling, Schema Design, Query Optimization.
- **Identity & Access Management:** SSO, RBAC, Federated Identity, Multi-Tenancy Security Patterns, IAM Validation.
- **Infrastructure as Code:** Terraform, CloudFormation, Environment Provisioning.
- **Messaging & Event-Driven Architecture:** Message Queues, Event-Driven Architecture, Middleware Integration Patterns, Webhooks.
- **Web Technologies:** Dotnet Core 6.0 Web API, HTML/HTML5, CSS2/CSS3, JavaScript, TypeScript, Angular, React JS.
- **CI/CD Tools:** Jenkins, Git, GitLab, Circle CI, TeamCity, Sause Labs.
- **Testing Tools:** Appium, Selenium WebDriver, Puppeteer, Lambda Testing Tool, Postman, JEST, Xcode, AVD.
- **Build Tools:** Yarn, NPM, NPX, Maven, ANT.
- **Browsers:** Google Chrome, Mozilla Firefox, MS Edge, Internet Explorer.
- **IDEs:** VS Code, Eclipse, Visual Studio, IntelliJ, PyCharm.
- **Version Control:** GitHub, GitLab, Bitbucket, SourceTree.
- **Add-Ons:** Selenium IDE, Selenium Builder, Gherkin, PgAdmin, ANDI, AXE, Confluence, Firebug.
- **Test Management Tools:** Jira, Azure DevOps, TestRail, Bugzilla.
- **Testing Phone OS:** iOS, Android, and Amazon.

EXPERIENCE

Test Automation Engineer

Oct 2024 – Present

PCMI LLC (Full Time)

Chicago, IL, USA

- Engineered and owned a **Test Automation** framework using **C#** with **Selenium**, architecting it to modern standards and driving **Playwright**-based UI automation across the PCRS SaaS platform.
- Directed execution of automated test cases for functional, integration, and regression testing, owning the platform's reliability and **production deployment** readiness under real usage conditions.
- Architected **CI/CD** pipeline integration for automated tests using **TeamCity**, driving continuous testing and rapid feedback across development sprints.
- Translated business requirements into test strategy and drove thorough test coverage across releases, partnering directly with developers, QA engineers, and product managers.
- Led validation of **RESTful API** authentication and authorization flows, including **OAuth2** and **API key** based access, owning end-to-end **API integration** testing for platform services.
- Owned **SQL** query design and maintenance for **database** testing and **data validation** across platform environments.
- Drove defect-free delivery of the PCMI internal platform, PCRS, by architecting thorough automated test coverage for core business workflows.

- Led defect triage, documenting and tracking issues found during automated testing while partnering with development teams to diagnose and resolve root causes.
- Drove test-result analysis, delivering actionable insights and recommendations to the development and **Engineering** team to guide release decisions.
- Extended **SaaS** testing ownership into **system integration testing**, UAT, and production validation across customer-facing modules.
- Owned data validation and backend verification during automated and manual test cycles, driving **data modeling** and **schema** consistency across the platform.
- Mentored newer QA team members on **test automation** architecture, **Agile/Scrum** ceremonies, and **API testing** techniques, raising overall test coverage quality.

Environments: Java · C# · Selenium WebDriver · Playwright · Appium · BDD (Cucumber) · TeamCity (CI/CD) · Jenkins · Jira · REST APIs · OAuth2 · API Key Management · SQL · Data Validation · TestNG · Postman · Agile/Scrum · SaaS Testing · System Integration Testing · UAT · Production Validation · Schema Validation

SDET / QA Automation Engineer

Dec 2023 – Sept 2024

Client: CVS

Remote, USA

- Owned detailed test plans, test cases, and scripts for functional and non-functional testing using **C#**, **TypeScript**, **.NET**, **Playwright**, and React Testing Library for CVS **healthcare** platform services.
- Drove comprehensive functional UI automation using **Playwright**, elevating the dependability and user experience of **payer-facing** and **healthcare** web applications.
- Partnered directly with developers and platform **Engineering** teams to architect automated test integration into the **CI/CD** pipeline, utilizing **GitLab** for version control and deployment gating.
- Owned **API** testing strategy with Postman and XUnit, validating **RESTful API** contracts, **OAuth2/JWT** token-based authentication, and **webhook** delivery for downstream integrations.
- Led testing of cloud-native applications on **AWS Lambda**, verifying **API Gateway** routing and **EventBridge**-triggered **event-driven** workflows for reliable performance in cloud environments.
- Architected **Cucumber**-based **BDD** test suites covering **prior authorization**, **utilization management**, claims and eligibility **healthcare data** flows for **health plans** and **managed care** customers.
- Drove **Selenium WebDriver** automation of web UI functionalities in CVS Health Care Systems, including **HIPAA**-regulated member data screens.
- Applied **Behavior-Driven Development (BDD)** practices to architect clean, well-structured **Java** code across the automation suite.
- Owned manual and automated RESTful API testing with Postman and **RESTAssured**, validating **SAML 2.0** and **mTLS**-secured service-to-service calls.
- Managed test case architecture with **JUnit** and **TestNG** frameworks, driving traceability between requirements and automated coverage.
- Partnered with cross-functional teams, including **Product** and Solutions teams, to define testing strategy and expand test coverage for **integration patterns** between internal and partner systems.
- Drove regression test automation to enhance efficiency and minimize manual effort across **SQL** and **NoSQL**-backed services, including **MongoDB**-based data stores.
- Led performance testing with tools like JMeter to assess system behavior under load conditions for **middleware** and **message queue**-driven components.
- Drove agile ceremonies and cross-functional standups with Engineering, Product, and Customer Success stakeholders to align priorities and maintain delivery flexibility.

Environments: Cucumber · JUnit · Behavior-Driven Development (BDD) · Selenium WebDriver · Postman API · RESTAssured · OAuth2 · JWT · SAML 2.0 · mTLS · Splunk · Swagger API · API Automation · Web Automation Testing · Gherkin · MySQL · MongoDB · AWS Lambda · API Gateway · EventBridge · Amazon Web Services (AWS) · Eclipse · IntelliJ IDEA · Git · GitLab · TestNG · Java · Python · Maven · TestRail · Jenkins · HIPAA · HL7 · Healthcare Data Standards · Prior Authorization · Utilization Management · Health Plans · Managed Care Organizations

Software Engineer

Sept 2021 – Dec 2022

Infor

Hyderabad, India

- Owned 90% **Test Case** development with **Selenium WebDriver** and Java Robot, architecting the Test Case framework around the modern **Page Object Model**.

- Engineered automation scripts, driving a 95% improvement in product health by aligning with requirements using **Appium** and SeeTest as part of Mobile testing.
- Architected **Cucumber**-based **BDD** test suites for enterprise application workflows.
- Delivered 80% proficiency in the high-level coding language XTEND, owning **data transformation** logic within the application layer.
- Owned mobile platform testing using JIRA, QTest, ACSS, validating **RESTful API** calls behind mobile app authentication (**OAuth2/JWT**) flows.
- Drove a 90% focus on **API Testing** using Postman and SoapUI, verifying **schema**-level request/response contracts.
- Led 80% of Product Design reviews with Product managers and Software Engineers, driving efficiency in test case development and deployment.
- Partnered with cross-functional teams through 2-week sprints and monthly releases, owning defect testing across an **n-tier application** architecture.
- Directed preparation of testing solutions, performed testing, and delivered test results across **integration** touchpoints between internal services.
- Owned test case, script, and plan development end-to-end, ensuring quality objectives were met for **SQL**-backed data layers.
- Led defect isolation and root-cause analysis, tracking issues to resolution in partnership with Engineering.
- Partnered with team members to assess technical challenges and higher-risk components and interfaces, including **middleware** integration points.
- Drove regression testing to ensure continued quality and functionality of applications across releases, validating backward-compatible **API** contracts.
- Owned QA metrics and KPI reporting for working projects, driving **data-driven** decisions for release readiness.

Environments: Automation Testing · Selenium · TestNG · Jest · Cucumber · XTEND · Android Studio · Java · TestRail · Jenkins · Appium · Manual Testing · Jira · Playwright · QTest · Test Planning · Test Cases · System Testing · Software Testing · REST APIs · OAuth2 · JWT · SQL · Schema Validation · n-Tier Architecture · Mobile API Testing

Associate Engineer QA

June 2019 – August 2021

Virtusa

Hyderabad, India

- Delivered 85% hands-on **Selenium WebDriver** automation using **Cucumber** within a **BDD Framework** for enterprise client engagements.
- Owned manual testing across Functional (80%), Integration (85%), System (90%), Regression (85%), and Unit Testing (80%).
- Engineered automated test cases for mobile applications using **Appium** with **Java**, driving defect identification across **API**-driven mobile workflows.
- Drove regression testing to ensure that new updates or changes didn't negatively impact existing functionalities or downstream **integration** points.
- Led participation in **Agile Methodology** and **Scrum** Teams, engaging in 90% of Agile projects and Scrum teams, and driving 80% of **CI/CD** pipelining tasks.
- Owned the testing process end-to-end, including Writing (90%), Executing (95%), Logging (85%), Tracking (80%), and Prioritizing (90%) defects.
- Delivered comprehensive test reports outlining discovered issues, resolutions, and recommendations to Engineering and Product stakeholders.
- Built working knowledge covering 75% of the technical stack in an **n-tier application**, including **SQL**-backed data layers.
- Owned validation of **RESTful API** responses and basic **authentication** flows as part of **system integration testing** for client-facing enterprise applications.
- Managed test data sets and configured test environments to support consistent execution of manual and automated test cycles across releases.
- Owned test procedure, defect log, and release sign-off documentation to support audit and knowledge-transfer needs for enterprise clients.
- Partnered with developers during code reviews to identify testability gaps and drive **API** and **data** layer test coverage improvements.

Environments: Java · J2EE · Appium · Selenium WebDriver · Cucumber · BDD · Production Testing · System Integration Testing · UAT · API Testing · REST APIs · Authentication Testing · SQL · n-Tier Architecture · Manual Testing · Regression Testing · Test Documentation · Agile/Scrum · CI/CD · Defect Tracking · JIRA

AWARDS & ACHIEVEMENTS

- 2019-11 Oracle Certified Associate, JAVA SE8 Programmer.
- ISTQB Certified Professional.
- Microsoft Technology Associate (MTA) – Database Administration Fundamentals, HTML- 5 Application Development Fundamentals.
- Awarded “Software Process Achievement” Award for my contribution towards the project in Virtusa.
- Received Customer Appreciation mails for quality releases.